

Multiple Choice (10 marks)

1. A water pipe in which water pressure is 80 lb/in.^2 springs a leak, the size of which is $.01 \text{ sq in.}$ in area. How much force is required of a patch to "stop" the leak?
 - (a) 8 oz
 - (b) 0.008 lb
 - (c) 8 kg
 - (d) 800 lb
 - (e) .8 lb
2. A nucleus of mass M emits a γ ray of energy E_γ . The nucleus was initially at rest. What is
 - (a) $E_\gamma^2 / 4Mc^2$
 - (b) $2E_\gamma^2 / Mc^2$
 - (c) E_γ / Mc^2
 - (d) $3E_\gamma^2 / 2Mc^2$
 - (e) $E_\gamma / 4Mc^2$
3. When 100 J are put into a device that puts out 40 J of useful work, the efficiency of the device is
 - (a) 25%
 - (b) 100%
 - (c) 60%
 - (d) 50%
 - (e) 75%
4. Find the minimum energy of a simple harmonic oscillator using the uncertainty principle.
 - (a) $\frac{h\omega}{4\pi}$
 - (b) $m\omega^2/2$
 - (c) $\frac{h}{m\omega^2}$
 - (d) $\frac{h^2}{2\pi m\omega}$
 - (e) $h^2/m\omega$
5. What is the total charge in coulombs of 75.0 kg of electrons?
 - (a) $-2.05 \times 10^{13} \text{ C}$

- (b) $-1.32 \times 10^{13} \text{C}$
- (c) $-1.98 \times 10^{13} \text{C}$
- (d) $-0.98 \times 10^{13} \text{C}$
- (e) $-1.44 \times 10^{13} \text{C}$

True / False (10 marks)

Answer True or False

6. True or False: An electron can reach a speed of $1.8 \times 10^7 \text{ m/s}$ by being accelerated across a potential difference of 100 V.
7. True or False: A uniformly charged conducting sphere with a diameter of 1.2 m and surface charge density of $2 \times 10^{-6} \text{ C/m}^2$ has a total charge of $2.5 \times 10^{-6} \text{ C}$.
8. True or False: An atom with filled $n = 1$ and $n = 2$ energy levels contains a total of 12 electrons.
9. True or False: The focal length of the lens is -20 inches, indicating that it is a diverging lens.
10. True or False: The specific heat capacity of sand is always equal to that of vegetation, leading to the same temperature change for the same amount of heat added.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the formation of the image by a bi-convex lens with a radius of curvature of 20 cm when an object of height 2 cm is placed 30 cm away. Discuss the characteristics of the image produced.
12. An observer moves past a meter stick at a velocity of half the speed of light. Discuss how relativistic effects influence the measured length of the meter stick and calculate the observed length.

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